

Experiment 3: House Price Analysis Using NumPy

Aim

To calculate the average sale price of houses having more than four bedrooms.

Algorithm

1. Import NumPy.
2. Create an array containing bedrooms, square footage, and sale price.
3. Select houses with more than four bedrooms.
4. Calculate their average sale price.
5. Display the results.

Code

```
import numpy as np

house_data = np.array([
    [3, 1200, 200000],
    [5, 1800, 350000],
    [4, 1500, 280000],
    [6, 2200, 450000],
    [5, 2000, 400000],
    [3, 1300, 220000]
])

selected_houses = house_data[house_data[:, 0] > 4]

average_price = np.mean(selected_houses[:, 2])

print("Selected Houses:")
print(selected_houses)
print("Average Sale Price:", average_price)
```

Input

House details containing the number of bedrooms, square footage, and sale price.

Output

```
===== RESTART: C:/Users/arunp/OneDrive/Documents/FDS/EXP 3.py =====  
Houses with more than four bedrooms:  
[[ 5 1800 350000]  
 [ 6 2200 450000]  
 [ 5 2000 400000]]  
Average Sale Price: 400000.0
```

Result

The average sale price of houses with more than four bedrooms was calculated as **400,000.**